

An AI-agent platform that actually worked — in 30 days

A Big Four professional-services firm — internal AI tooling

THE STAKES

The firm wanted an internal platform where its own teams could build AI agents for their workflows. Not one agent — an agent factory: an architect agent decomposes the request, specialized agents are created dynamically, a verifier agent checks the output, and a human approves at the gates. A bare-bones skeleton existed; nothing worked. Builds of this shape are routinely quoted in months, with teams of five or more — and most of what enterprises see of “agentic AI” is demos that collapse on contact with real requirements.

WHAT I DID

Two engineers total — my team lead and me. We took the skeleton to a **working multi-agent platform in 30 days**:

- Dynamic agent creation with human-in-the-loop controls at every gate
- Architect / verifier role separation — the work is produced, then independently checked, before a human ever signs off
- Delivered on Google’s stack: Vertex AI, Gemini, and the ADK agent framework

OUTCOMES

- Delivered on the 30-day deadline, demoed to stakeholders, accepted for internal use by the firm’s teams.
- Two engineers and thirty days, against the multi-month larger-team builds this class of system usually gets quoted at. (Adoption breadth beyond delivery is the client’s data — no claims made.)

WHY IT MATTERS

This is the reference for AI systems that survive enterprise requirements: verification gates and human approval designed in from day one, not patched on when the pilot fails review. The same discipline runs through every AI-assisted delivery Alpine does — agents produce the work, verification proves it.

STACK

Vertex AI · Gemini · Google ADK · Python

This is the engineering standard behind Alpine’s fixed-price delivery audit: findings with file-and-line evidence, an implementation playbook, and an execution-ready backlog your team can run with. References on request.

Stanislav Petrov · Alpine · slavpetroff@gmail.com · linkedin.com/in/stanislav-petrov-a72739105